

Sensors & Systems

Kalman Filter for Nonlinear Systems — EKF & UKF

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`np.linalg.eigh((n + λ)P)` decomposes the scaled covariance matrix into:

- `eigvals` = Λ — how much variance exists in each direction
- `eigvecs` = V — which directions the variance points in

Think of it as finding the natural axes of the uncertainty ellipse and how stretched it is along each axis.

`S = eigvecs @ np.diag(np.sqrt(eigvals))` builds the matrix square root

$$S = V\sqrt{\Lambda}$$

so each column of S points in an eigenvector direction (shape of uncertainty) scaled by $\sqrt{\lambda_i}$ (how far to push in that direction). Together $S \approx \sqrt{(n + \lambda)P}$, giving the sigma point offsets from the mean.

`np.maximum(eigvals, 0)` clips any negative eigenvalues to zero before taking the square root. P should be positive definite by construction, so all eigenvalues should be > 0 , but floating-point accumulation over many steps can produce tiny negative values. Taking $\sqrt{\text{negative}}$ would give NaN and silently break the filter — the clip prevents this.